YIFAN ZHENG, PHARM.D.

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Professional Summary: Ph.D. candidate and Pharm.D. specializing in pharmacoepidemiology, real-world evidence (RWE), and health outcomes research. Experienced in observational and cohort study design, causal inference, drug safety and pharmacovigilance, comparative effectiveness research, and longitudinal modeling of electronic health record (EHR), pharmacy claims, and clinical-trial data. Skilled at translating real-world data into decision-relevant evidence for treatment outcomes, medication safety, and value demonstration.

SKILLS

Pharmacoepidemiology & Outcomes Research: Real-world evidence; observational and cohort study design; new-user and active-comparator designs; comparative effectiveness research; medication adherence and exposure measurement; drug safety and pharmacovigilance (FDA FAERS, WHO VigiBase); evidence synthesis and meta-analysis; patient-reported and economic outcomes

Quantitative Methods: Causal inference, survival analysis, regression and longitudinal modeling, temporal deep learning, machine learning, model-based and network meta-analysis, exposure–response modeling, study power and bias assessment

Data & Programming: R, Python, SQL, Git; large EHR and pharmacy-claims data, RxNorm / NDC / ICD-10 coding, FDA FAERS and WHO VigiBase, natural language processing, GPT / LLM APIs, reproducible analytic workflows

EDUCATION

University of Michigan — **Ann Arbor, MI**

Ph.D. Candidate, Clinical Pharmacy Translational Science — Health Services Research Track — *Sep 2022 – Dec 2026*
Joint Ph.D. Candidate in Scientific Computing, College of Engineering — *Sep 2022 – Dec 2026*
Dual Master of Science (M.S.) Candidate in Bioinformatics, Medical School — *Sep 2024 – Dec 2026*
Doctor of Pharmacy (Pharm.D.), with Distinction, College of Pharmacy — *Sep 2016 – May 2020*

China Pharmaceutical University (CPU) — **Nanjing, China**

Bachelor of Science in Pharmacy (The National Bio-science Honor Program) — *Sep 2012 – Jun 2016*

WORK EXPERIENCE

Real-World Evidence Analyst (Contractor at AbbVie) — HireGenics, Inc. — **Remote, USA**
Real-World Evidence Group, AbbVie — *Oct 2025 – Present*

- Synthesize large-scale real-world oncology data to characterize treatment and care gaps in ovarian cancer, informing evidence generation and medical strategy decisions.
- Build machine-learning models to segment providers by engagement opportunity, supporting targeted Medical Affairs strategy and efficient deployment of field resources.
- Launched and lead an internal AI-driven analytics initiative, training colleagues on AI-assisted workflows and improving analytic efficiency across the team.

Quantitative Pharmacology Co-op (Mentor: Dan Li, Ph.D.; Manager: Islam Younis, Ph.D.) — **West Point, PA**
Merck & Co. — *Feb 2025 – Aug 2025*

- Conducted model-based network meta-analysis (MBMA) and evidence synthesis across ophthalmology clinical trials to evaluate comparative efficacy, safety, and effectiveness, supporting dose selection, go/no-go decisions, and competitive positioning.
- Applied nonlinear mixed-effects and random forest models to quantify exposure–response and comparative treatment effects.
- Used AI-assisted evidence-extraction platforms (Covidence, Elicit) and collaborated with IT to implement NLP-driven systematic literature search; completed internal NCA training and led a modeling study group.

Clinical Pharmacist & Lecturer (PI: Kejing Tang, M.D., Ph.D.) — **Guangzhou, China**
The First Affiliated Hospital of Sun Yat-sen University — *Jul 2020 – Aug 2022*

- Delivered clinical pharmacy services across 3 units (150+ patients each) and developed a new outpatient lung-cancer service using a medication-management app, improving patient medication adherence.
- Supervised two master's students to two successful publications; taught a therapeutics course to 30 students, earning a teaching award and recording national exemplary lectures.

RESEARCH EXPERIENCE

Medication Data Science Lab (PI: Corey Lester, Pharm.D., Ph.D.) — **Ann Arbor, MI**
Department of Clinical Pharmacy Translational Science, University of Michigan — *Sep 2022 – Present*

- Lead a dissertation developing pharmacoepidemiologic and machine-learning methods for medication safety and effectiveness across EHR, insurance, and pharmacy-claims data.
- Quantified how prescription, dispensing, and sold data sources change real-world medication adherence and exposure measures, with direct implications for RWE study design (first author, *JMCP* 2026; Top 5% Research Presentation Award, ISPOR 2026).
- Built temporal deep-learning models on longitudinal pharmacy-fill sequences to predict cardiovascular disease risk in real-world cohorts.
- Applied causal machine learning and survival analysis to estimate individualized effects of medication adherence on blood-pressure outcomes in hypertension.
- Engineered Python / GPT-4 and RxNorm pipelines to analyze 100K+ prescription and dispensing records for medication-exposure measurement and dispensing-error detection; mined millions of e-prescription change requests in R to characterize pharmacist

interventions and cost outcomes.

Lung Cancer & Pharmacy Lab (PI: Kejing Tang, M.D., Ph.D.) **Guangzhou, China**
The First Affiliated Hospital of Sun Yat-sen University *Jul 2020 – Aug 2022*

- Led pharmacoepidemiology and pharmacovigilance studies of lung-cancer immunotherapy using FDA FAERS and WHO VigiBase data, characterizing comparative adverse-event profiles across treatment regimens.
- Assessed the quality and readability of OTC medication inserts and online health information using novel quantitative methods, raising awareness of patient-facing drug-information gaps.

Pharmacy Informatics Research Lab **Ann Arbor, MI**
Department of Clinical Pharmacy Translational Science, University of Michigan *Oct 2018 – Apr 2020*

- Designed and led a retrospective cohort study of workload, readability, quality, and safety of over one million outpatient pharmacy transcriptions of patient directions in the U.S. (published in *BMJ Quality & Safety*).

SELECTED PUBLICATIONS

First-author or co-author of 25+ peer-reviewed publications (Google Scholar); selected pharmacoepidemiology work below.

1. **Zheng Y**, Farris KB, Coe AB, Dorsch MP, Najarian K, Lester CA. The impact of data source on real-world medication adherence and exposure measures: from prescription to sold. *Journal of Managed Care & Specialty Pharmacy*. 2026;32(4):434–444.
2. **Zheng Y**, Gong J, Lester CA. Evaluating the impact of pharmacy-led RxChange interventions on medication use and cost outcomes for electronic prescriptions. *Journal of the American Pharmacists Association*. 2025;65(3):102349.
3. **Zheng Y**, Whitaker M, Geer A, Gong J, Lester C. Comparing the performance of a large language model and RxNorm mapping for detecting medication dispensing errors from real-world data. *2025 IEEE International Conference on Healthcare Informatics (ICHI)*. 2025:98–106.
4. Li H†, **Zheng Y†**, Xu P, Li Z, Kuang Y, et al. Comparison of pneumonitis risk between immunotherapy alone and in combination with chemotherapy: an observational, retrospective pharmacovigilance study. *Frontiers in Pharmacology*. 2023;14:1142016. (†co-first authors)
5. **Zheng Y**, Jiang Y, Dorsch MP, Ding Y, Vydiswaran VGV, Lester CA. Work effort, readability, and quality of pharmacy transcription of patient directions from electronic prescriptions: a retrospective observational cohort analysis. *BMJ Quality & Safety*. 2021;30(4):311–319. *(Top Article Award)*

SELECTED PRESENTATIONS

1. **Zheng Y**, Lester C. When adherence estimates differ: the impact of data source and gap handling in real-world medication-use assessment. *ISPOR (International Society for Pharmacoeconomics and Outcomes Research) 2026 Annual Meeting*, Philadelphia, PA. May 2026. [*Top 5% Finalist, Research Presentation Award; Poster Tour Presentation*]
2. **Zheng Y**, Lester CA. Leveraging real-world medication-taking behaviors to support personalized treatment decisions in patients with hypertension. *Midwest Social and Administrative Pharmacy Meeting*. Jun 2024. [Oral — *Audience Choice Award*]
3. **Zheng Y**, Alexander G, Chawla A, Li D. Model-based meta-analysis to support drug development for diabetic macular edema. *ASCPT Annual Meeting*. Mar 2026. [Poster — *Presidential Trainee Award; Top Poster Ribbon*]
4. **Zheng Y**, Rowell B, Chen Q, et al. Human-centered AI for preventing medication dispensing errors: a focus group study. *AMIA Annual Symposium*. Nov 2023. [Poster]

HONORS & AWARDS

Rackham Predoctoral Fellowship — a top dissertation award of the University of Michigan	Mar 2026
Research Presentation Award (Top 5% Finalist) — ISPOR Annual Meeting	May 2026
Presidential Trainee Award — American Society for Clinical Pharmacology and Therapeutics (ASCPT)	Mar 2026
Rho Chi Honor Society — University of Michigan College of Pharmacy	Mar 2025
Audience Choice Award — Midwest Social and Administrative Pharmacy Meeting, University of Iowa	Jun 2024
Phi Kappa Phi Honor Society (Top 10%) — University of Michigan	Nov 2023
Top Article Award — BMJ Quality & Safety	Mar 2021
Thomas D. Rowe Award (service, leadership & academic excellence) — University of Michigan College of Pharmacy	Apr 2020
National Scholarship (full tuition & stipend, ~$290K) — China Scholarship Council	2016 – 2020
Presidential Scholarship (University's Highest Honor) — China Pharmaceutical University	Jan 2014

LEADERSHIP & PROFESSIONAL SERVICE

Founder & Inaugural President — ISPE Student Chapter, University of Michigan	Aug 2025 – Present
Co-President — Annappella (a cappella club), University of Michigan	Apr 2024 – Present
Graduate Student Committee Member — University of Michigan	Jan 2023 – Jan 2025
Graduate Student Instructor — CPTS 832: Pharmacy Informatics, University of Michigan	Jan 2023 – Apr 2024
Journal Peer Reviewer — 16 review records across medical informatics and pharmacy journals	2022 – Present
Member — ISPE; American Society for Clinical Pharmacology and Therapeutics (ASCPT)	2025 – Present